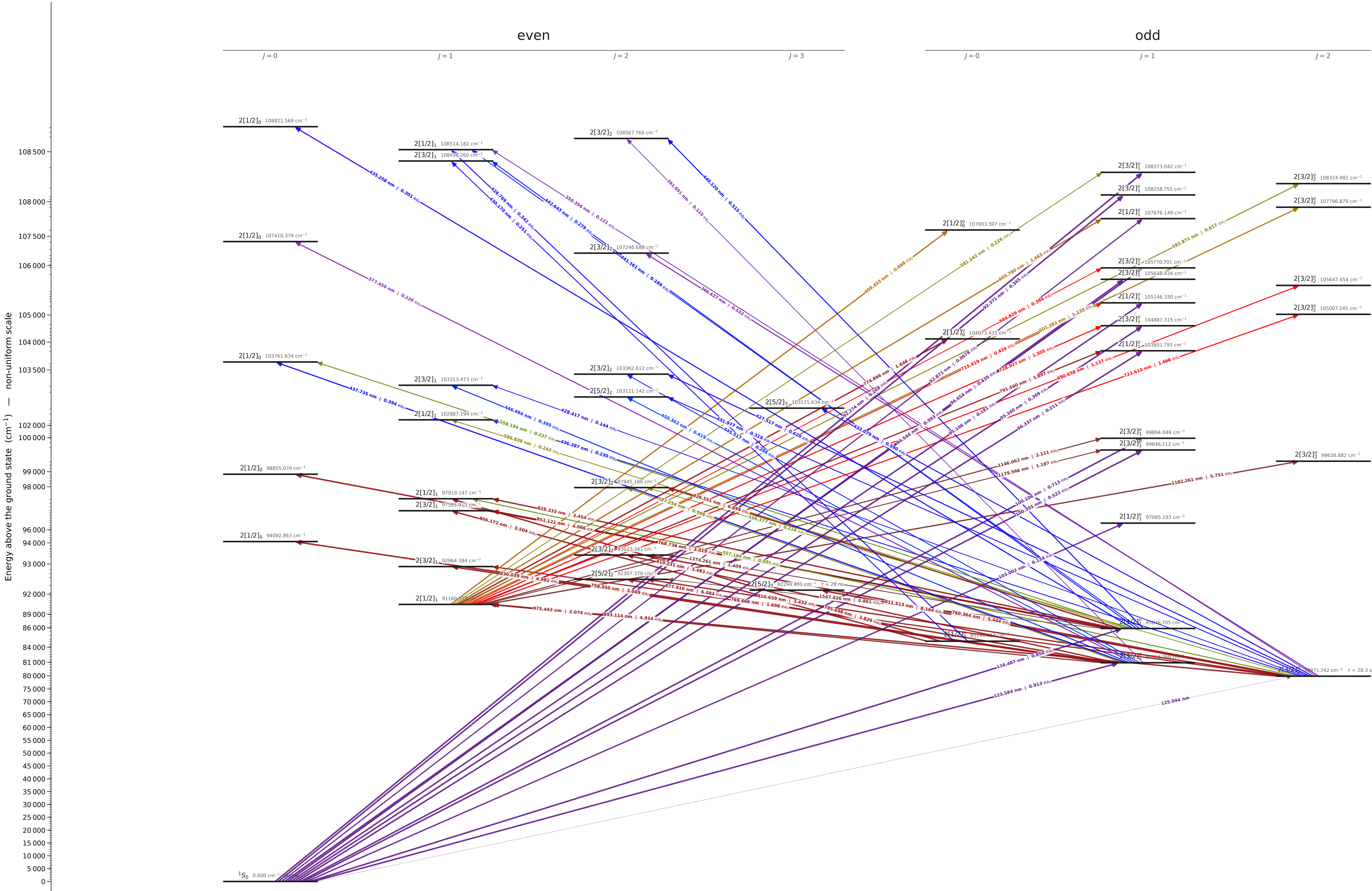
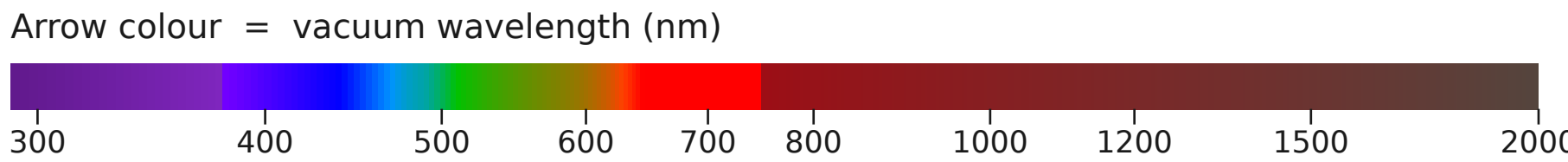


Krypton

Kr (Kr I). Strongest classified transitions below 2 μm (NIST): 45 levels, 71 lines.

Kr⁺ ionisation limit 112914.43 cm⁻¹ = 13.99961 eV (88.563 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|(J||e\mathbf{r}||J')|$ in ea0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J e\mathbf{r} J' \rangle $ (ea0)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
¹ S ₀ – 2[3/2] ₂ ^o	125.0442	–	2397.4925	–	3.540e-02	5.624e-09	100	125.0 nm M2 decay of the metastable 3P2 state
2[3/2] ₂ ^o – 2[5/2] ₃	811.5131	811.2900	369.4240	8.164	3.610e+07	5.684	100	811.3 nm
2[3/2] ₂ ^o – 2[3/2] ₂	819.2308	819.0056	365.9438	3.483	8.940e+06	–	–	819 nm second step of the optical

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

⁸³Kr hyperfine structure (I = 9/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2[3/2] ₂ ^o	-244.04(3)	-453.4(6)	F=5/2: +2476.65 F=7/2: +1820.86 F=9/2: +864.35 F=11/2: -477.87 F=13/2: -2309.70
2[1/2] ₁	-143.00(9)	–	F=7/2: +786.50 F=9/2: +143.00 F=11/2: -643.50
2[5/2] ₃	-103.910(20)	-437.0(7)	F=3/2: +1514.21 F=5/2: +1345.48 F=7/2: +1083.77 F=9/2: +703.58 F=11/2: +172.14 F=13/2: -550.62 F=15/2: -1512.04

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
2[3/2] ₂ ^o – 2[5/2] ₃	811.513	83-78	201.33(13)

Shift = ν(first isotope) – ν(second isotope).